



CAHSI LOCAL REU RESEARCH PLAN

SEMESTER: **SPRING** YEAR: **2026**

Start Date: **1/26/2026** End Date: **4/10/2026**

Project Title: An Analysis and Simulation of Defending Tracking Control Neural Network for Known Affine Discrete-Time Nonlinear Systems Against Adversarial Attacks	Mentor Name Pavlo Tymoshchuk Institution <i>University of North Texas</i>	Student Name Ablasse Kingcaid-Ouedraogo Institution Texas A&M University
Problem Description: [Describe problem being addressed and relationship to broader problem] Overview of existing methods of defending tracking control against adversarial attacks will be performed. Based on obtained results the most efficient approaches of the defense will be selected. The selected methods will be applied to defending tracking control neural network (NN) for known affine discrete-time nonlinear systems against adversarial attacks. For this purpose, the tracking control NN will be described by the difference equation with variable structure. The existence and uniqueness of the correct steady states of the state variable trajectories of the difference equation in the presence of adversarial attacks will be analyzed and shown that they exist and are unique. The stability and convergence to the steady states of the state variable trajectories of the difference equation in the presence of adversarial attacks will be investigated and shown that the trajectories are globally stable and globally convergent to the correct steady states in a finite number of steps. A functional block-diagram of the tracking control NN and its defending part will be presented and analyzed. The diagram will be implemented in MATLAB and/or Python and/or C++ and/or Compute Unified Device Architecture (CUDA). These implementations will be compared with each other. The functional block-diagram will be compared with other competitive analogs and shown that it has less computational and hardware implementation complexity. Computer simulations confirming theoretical derivations and illustrating operation of the defended tracking control NN for known affine discrete-time nonlinear systems will be fulfilled.		
Research Significance: [Why should people care?] The research is significant since tracking control neural network for known affine discrete-time nonlinear systems that is used in many useful applications needs methods of defending against adversarial attacks.		
Research Goals and Objectives: [Alternative: Research Questions] [Short and long term goals] The project has the following goals: Implement in MATLAB and/or Python and/or C++ and/or Compute Unified Device Architecture (CUDA) a functional block-diagram of the tracking control NN and its defending part; compare a performance of the implementations with each other and those of other competitive analogs. The project objectives are: Describe the tracking control NN by the difference equation with variable structure; analyze the existence and uniqueness of the correct steady states of the state variable trajectories of the difference equation in the presence of adversarial attacks and show that they exist and are unique; investigate the stability and convergence to the steady states of the state variable trajectories of the difference equation in the presence of adversarial attacks and show that the trajectories are globally stable and globally convergent to the correct steady states in a finite number of steps; present and analyze a		



functional block-diagram of the tracking control NN and its defending part; implement the diagram in MATLAB and/or Python and/or C++ and/or Compute Unified Device Architecture (CUDA); compare these implementations each other; compare the functional block-diagram with other competitive analogs and show that it has less computational and hardware implementation complexity; fulfill computer simulations confirming theoretical derivations and illustrating operation of the defended tracking control NN for known affine discrete-time nonlinear systems.

Assigned tasks	Start Date	Deadline
Describe the tracking control NN by the difference equation with variable structure based on close analogs presented in [1] - [3].	1/26/2026	2/2/2026
Analyze the existence and uniqueness of the correct steady states of the state variable trajectories of the difference equation in the presence of adversarial attacks and show that they exist and are unique.	2/2/2026	2/9/2026
Investigate the stability and convergence to the steady states of the state variable trajectories of the difference equation in the presence of adversarial attacks and show that the trajectories are globally stable and globally convergent to the correct steady states in a finite number of steps.	2/9/2026	2/16/2026
Present and analyze a functional block-diagram of the tracking control NN and its defending part.	2/16/2026	3/2/2026
Implement the diagram in MATLAB and/or Python and/or C++ and/or Compute Unified Device Architecture (CUDA).	3/2/2026	3/9/2026
Prepare and submit to the CAHSI Local REU Program midterm report.	3/9/2026	3/16/2026
Compare the implementations of the diagram in MATLAB and/or Python and/or C++ and/or Compute Unified Device Architecture (CUDA).	3/16/2026	3/23/2026
Compare the functional block-diagram with those of other competitive analogs and show that it has less computational and hardware implementation complexity.	3/23/2026	3/30/2026
Fulfill computer simulations confirming theoretical derivations and illustrating operation of the defended tracking control NN for known affine discrete-time nonlinear systems. Compare the simulation results with those of other competitive analogs.	3/30/2026	4/6/2026
Prepare and submit to the CAHSI Local REU Program research poster and final report.	4/6/2026	4/10/2026
Present the research poster at the Annual GMIS Conference.	9/17/2026	9/19/2026
Deliverables/Milestones	Date Due	
Description of the tracking control NN by the difference equation with variable structure based on close analogs presented in [1] - [3].	2/2/2026	



Description of existence and uniqueness analysis of the correct steady states of the state variable trajectories of the difference equation in the presence of adversarial attacks.	2/9/2026
Description of analysis of stability and convergence to the steady states of the state variable trajectories of the difference equation in the presence of adversarial attacks.	2/16/2026
Description of functional block-diagram of the tracking control NN and its defending part.	3/2/2026
Description of the diagram implementation in MATLAB and/or Python and/or C++ and/or Compute Unified Device Architecture (CUDA).	3/9/2026
The midterm report.	3/16/2026
Description of comparison of the diagram implementations in MATLAB and/or Python and/or C++ and/or Compute Unified Device Architecture (CUDA).	3/23/2026
Description of comparison of the functional block-diagram with those of other competitive analogs.	3/30/2026
Description of the computer simulation results of the defended tracking control NN for known affine discrete-time nonlinear systems and their comparison with those of other competitive analogs.	4/6/2026
The research poster and final report.	4/10/2026
Presentation of the research poster at the current Annual GMiS Conference.	9/17/2026 - 9/19/2026

References

- [1] P. Tymoshchuk, "A tracking control neural network for the known affine in the inputs discrete-time nonlinear systems," in Wybrane problemy sterowania, napędu oraz układów zasilania; Selected problems of control, drives, and power systems, D. Grabowski, Ed., Gliwice, Poland: Silesian University of Technology Publishing House, 2024, pp. 39–49.
- [2] P. Tymoshchuk, "Neural network optimal control for discrete-time nonlinear systems with known internal dynamics," Neural Computing and Applications, vol. 36, pp. 4421–4434, Mar. 2024. <https://doi.org/10.1007/s00521-023-09244-y>.
- [3] M. B. Jedh, L. B. Othmane, and A. K. Somani, "Improvement and evaluation of resilience of adaptive cruise control against spoofing attacks using intrusion detection system," in Proc. Risks and Security of Internet and Systems: 18th International Conference, Rabat, Morocco, 2023, pp. 51–66, <https://doi.org/10.48550/arXiv.2302.00876>.